

Pardha Sai Krishna Ala

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PROFILE

Doctoral candidate in Automotive Engineering at Clemson University, with a research focus on vehicle parameter estimation for haptic-based teleoperation. Experience in autonomous vehicle simulation, vehicle dynamics modeling, control strategy development, and graduate teaching.

EDUCATION

Clemson University – CU-ICAR Aug 2021 – Present

Doctor of Philosophy Candidate / GPA: 3.83/4.00

Research focus: Estimation of vehicle parameters for haptic-based teleoperation.

Clemson University – CU-ICAR Aug 2020

M.S. in Automotive Engineering / GPA: 3.52/4.00

Major: Vehicle Dynamics.

National Institute of Technology Calicut May 2016

B.S. in Mechanical Engineering / GPA: 7.85/10.00

Calicut, Kerala, India.

RESEARCH EXPERIENCE

Research Lab Assistant, OpenCAV Lab Feb 2020 – Jul 2021

Clemson University / Advisor: Dr. Venkat Krovi

- Simulation testing and verification of scenarios for Autonomous Vehicles
- Design and development of control strategies for Adaptive Cruise control for platooning strategies
- MATLAB, Simulink, and Metamoto were used to create Vehicle Dynamics models and tested for simulations.
- Use of Open-source software like Carla, LGSVL and Autoware are used to perform AV simulations.

TEACHING EXPERIENCE

Graduate Teaching Assistant Fall 2021 & Fall 2022

Dynamic Performance of Vehicles / Professor: Dr. Matthias J. Schmid

- Development of graduate project: Modeling of rear steered vehicle and its simulation for different trajectories for varying fidelity models
- Development of assignments and offering office hours for undergraduate and graduate students
- Offering classes on Transport theorem, Vertical dynamics

PROFESSIONAL EXPERIENCE

Head of Design and Development

Sep 2016 – Apr 2018

Uravu Manufacturers Pvt. Ltd.

- Contributed to the research, development, and testing of atmospheric water generator (AWG) models that convert atmospheric water vapor into potable water under different operating conditions.

Trainee Engineer

May 2016 – Aug 2016

ZF Wind Power Pvt. Ltd.

- Supported wind turbine gearbox manufacturing and quality control efforts to reduce heat-treatment defects.

PUBLICATIONS

1. **Pardha Sai Krishna Ala**, Ameya Salvi, Venkat Krovi, and Matthias Schmid. “Physics constrained learning of stochastic characteristics.” *IFAC-PapersOnLine*, vol. 59, no. 30, pp. 383–388, 2025.
DOI: 10.1016/j.ifacol.2025.12.267 Preprint Code
2. Ameya Salvi, **Pardha Sai Krishna Ala**, Jonathon M. Smereka, Mark Brudnak, David Gorsich, Matthias Schmid, and Venkat Krovi. “Online identification of skidding modes with interactive multiple model estimation.” *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025.
Preprint: arXiv:2409.20554 Code

SKILLS

Engineering software: MATLAB (Intermediate), Simulink (Intermediate), dSPACE (Basic), CarSim (Intermediate), SolidWorks (Intermediate).

Programming and robotics: C++ (Basic), Python (Intermediate), ROS.

Languages: English (Proficient), Telugu (Proficient), German (A2).

COURSEWORK

Automotive Systems: An Integrated overview • Project Management • Automotive Electronics • Perception and Intelligence • Automotive Manufacturing systems • Vehicle Dynamics • Automotive stability and control • Noise, Vibration, and Harshness • Tire Dynamics • Powertrain controls • On Board Diagnostics • Analysis of Linear Systems • Autonomy: Mobility, and Manipulation • Autonomy Systems and Science • Introduction to Data Science, ML, and Search Algorithms